

Semi-Supervised Fake News Detection with Mixture of Experts

Zhenyu Yang
School of Computing
Macquarie University
Sydney, NSW, Australia
zhenyu.yang@mq.edu.au

Chaoyu Yang*
School of Artificial
Intelligence
Anhui University of
Science and Technology
Huainan, Anhui, China
yangchy@aust.edu.cn

Wenfeng Xu
School of Artificial
Intelligence
Anhui University of
Science and Technology
Huainan, Anhui, China
2024201829@aust.edu.cn

Xiuxiu Hao
School of Artificial
Intelligence
Anhui University of
Science and Technology
Huainan, Anhui, China
2024201819@aust.edu.cn

Huan Wang
School of Computing and
Information Technology
University of Wollongong
Wollongong, NSW
Australia
hw226@uowmail.edu.au

Ge Zhang
School of Information and
Intelligent Science
Donghua University
Shanghai, China
gezhang@dhu.edu.cn

Xiaoxiao Ma
Oracle Corporation
Sydney, NSW, Australia
xiaoxiao.ma@oracle.com

Jun Shen
School of Computing and
Information Technology
University of Wollongong
Wollongong, NSW
Australia
jshen@uow.edu.au

Abstract

Single-expert fake news detectors, such as Graph Neural Networks (GNNs) and Large Language Models (LLMs), increasingly struggle to counter the diversifying camouflage tactics of modern adversaries, which range from semantic (e.g., mimicking writing styles) to structural (e.g., manipulating propagation paths). To address this, existing methods attempt to build a hybrid model by sequentially incorporating GNNs and LLMs; however, such hybridization blurs the distinction between experts and prevents critical cross-validation. In addition, existing methods rely heavily on vast labeled data, which is costly to acquire, particularly for fake news samples. In this paper, we propose a Semi-supervised Mixture of Experts framework for Fake news detection, namely S^2MOE-F . The core idea of S^2MOE-F is to establish a robust defense against multifaceted camouflage by cross-validating the complementary judgments of two independent experts, GNN and LLM. On the one hand, S^2MOE-F drives experts' judgments by using a One-Class Classification (OCC) objective, which constrains true news within a compact hypersphere and identifies samples outside this boundary as fake, reducing reliance on scarce fake news labels. On the other hand, S^2MOE-F generates high-confidence pseudo-labels based on consensus or divergence between experts to exploit abundant unlabeled data. In addition, we propose a novel reinforcement learning (RL)-based routing policy that dynamically determines the dominant expert for input samples without explicit supervision. Finally, we design a disentangled masked Transformer to ensure experts' specialization by reducing inter-expert redundancy. Extensive experiments on real-world datasets sourced from Web platforms and social media demonstrate the superior performance of S^2MOE-F .

*Corresponding author.



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ACM ISBN 979-8-4007-2307-0/2026/04
<https://doi.org/10.1145/3774904.3792312>

CCS Concepts

• Information systems → Data mining.

Keywords

Fake News Detection, Mixture of Experts, LLM, GNN

ACM Reference Format:

Zhenyu Yang, Chaoyu Yang, Wenfeng Xu, Xiuxiu Hao, Huan Wang, Ge Zhang, Xiaoxiao Ma, and Jun Shen. 2026. Semi-Supervised Fake News Detection with Mixture of Experts. In *Proceedings of the ACM Web Conference 2026 (WWW '26)*, April 13–17, 2026, Dubai, United Arab Emirates. ACM, New York, NY, USA, 12 pages. <https://doi.org/10.1145/3774904.3792312>

Resource Availability:

The source code of this paper has been made publicly available at <https://doi.org/10.5281/zenodo.18344931>.

1 Introduction

In the era of rapid dissemination of information within social networks, the widespread proliferation of fake news has become a critical concern. For example, the fake news of COVID-19 - the 'consuming highly concentrated alcohol can disinfect the body and kill the virus' - led to severe social damage, including but not limited to more than 5,800 hospitalizations [13]. Therefore, developing effective fake news detection algorithms is essential to ensure a more reliable web and protect the public from misinformation. In response, a variety of advanced approaches for fake news detection have emerged, primarily centered on Graph Neural Networks (GNNs) [2, 52, 53] and Large Language Models (LLMs) [12, 43, 44].

To counter these two mainstream detection technologies, fake news adversaries have correspondingly developed targeted evasion strategies, as shown in Fig. 1, including: **1) Semantic camouflage:** fake news adversaries could mimic the writing style of true news, fabricating "citation sources", and even relying on LLMs to craft textually indistinguishable misinformation [1]. This disguised information poses a substantial threat to semantic-based LLM detectors [5]. **2) Structural camouflage:** on the other hand, fake news adversaries could manipulate propagation paths through bots,

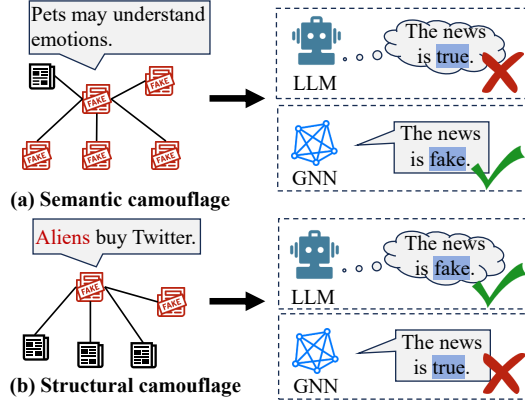


Figure 1: Two primary types of camouflage: (a) Semantic: The content is crafted to be plausible, but its propagation structure can reveal its malicious nature (e.g., by linking to other fake news). **(b) Structural:** The propagation structure is manipulated to appear benign, but its malicious nature can be revealed by semantic fallacies (e.g., "aliens").

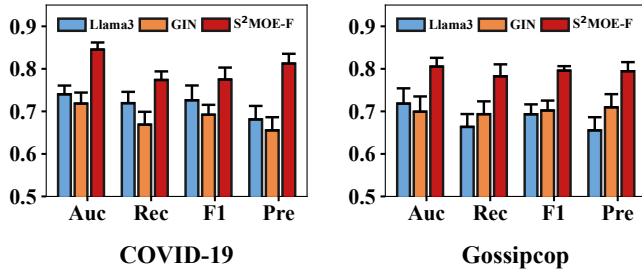


Figure 2: Performance comparison of S²MOE-F against single-expert on two datasets under a semi-supervised setting.

paid posters, and organized forwarding [26], thereby disrupting the structure of social networks. Because GNN-based detectors are implemented by analyzing network structures, they might be easily misled by deliberately manipulated propagation paths with structural camouflage [16]. Thus, single-expert evidence is prone to bias or deception. The final verdict on suspicious news requires cross-validating both the semantic and structural evidence.

Several very recent studies have enhanced LLMs' embeddings with GNNs to advance detection performance [15, 22, 23, 47]. However, the primary limitation of these methods is that they dissolve one expert of evidence into the other by treating LLM embeddings as simple node features for a GNN, which overlooks each expert's role as a distinct line of defense against semantic or structural camouflage. Such a fusion blurs the distinction between their judgments and prevents the critical cross-validation required to form a precise verdict [51]. In contrast, the Mixture-of-Experts (MoE) emerges as a well-suited concept specifically designed to integrate insights from diverse, specialized experts. Unfortunately, existing MoEs are not considered a robust defense against both structural and semantic camouflage. For instance, MIMoE-FND [19] and MisD-MoE [18] confine MoEs to news content (e.g., text and images), lacking critical

cross-validation against both semantic and structural camouflage. Furthermore, a fatal limitation of all the aforementioned approaches is their reliance on supervised learning, which necessitates vast amounts of labeled data. The extensive manual effort required to annotate these labels could be costly and labor-intensive, as exemplified by the CREDBANK dataset [25], where over 60 million tweets for 1,049 events were each annotated by 30 human evaluators. Thus, an innovative MoE framework is desired that can perform cross-validation against both structural and semantic camouflage while effectively training with a small portion of labeled data and large amounts of unlabeled data.

To build such a framework, several key challenges still lie in: **C1) Mitigating the severe redundancy between GNN and LLM experts.** The core MoE principle of expert specialization, typically achieved by assigning experts to distinct data modalities like text and images [18, 19], is fundamentally compromised when using GNN and LLM experts that both rely on a shared textual foundation. This shared reliance, where the same text serves as the LLM's direct input and the source for the GNN's node features, inevitably leads to significant redundancy between experts. **C2) Training the MoE framework under realistic semi-supervised fake news detection.** MoEs' reliance on abundant labeled data to jointly train multiple experts and gating networks severely limits their practical application [42]. This limitation is acute in fake news detection, as labeled news is not only inherently scarce but also costly and labor-intensive to obtain, especially for fake news samples. Consequently, the crucial and unexplored research problem is to develop an MoE framework that not only learns with high efficiency from the limited labeled data but also leverages the diverse insights of its experts to exploit the wealth of unlabeled data. **C3) Developing an expert routing mechanism with limited supervision.** In traditional MoEs, a gating network learns a routing policy from abundant labeled data, allowing the final decision to be dominated by the most suitable expert [36], e.g., the LLM for inputs with structural camouflage and the GNN for semantic camouflage. However, when such supervisory signals become insufficient, the traditional gating mechanism fails to learn this nuanced routing logic. Such a failure creates a crucial need for a new routing mechanism capable of learning effectively under limited supervision.

To address the above challenges, this paper proposes a Semi-supervised Mixture of Experts framework for Fake news detection, namely S²MOE-F¹. To **mitigate inter-expert redundancy (C1)**, we design a disentangled masked transformer module, which disentangles each expert's embedding into a set of distinct sub-embeddings. Subsequently, a learnable masked matrix is optimized via a covariance objective that minimizes redundancy between sub-embedding pairs from the different experts, thereby enforcing expert specialization. To enable **learning from limited labels (C2)**, we employ a One-Class Classification (OCC) objective, which constrains true news into a compact hypersphere and thereby identifies any sample falling outside this boundary as fake. This objective is highly suitable for realistic semi-supervised fake news detection, given the scarcity and high cost of labeled fake news examples. Meanwhile, we further **exploit unlabeled data (C2)** through a pseudo-labeling process, where experts' consensus or divergence is

¹Our code is available at <https://github.com/double-No1/S-2MOE-F>.

used to generate high-confidence pseudo labels for iterative training. To address the challenge of **routing experts with limited supervision (C3)**, we introduce a novel reinforcement learning (RL)-based router, which generates reward signals from the distance between the embedding of each expert and the center of the OCC hypersphere, thus dynamically assigning the dominant expert for input samples. Finally, to counteract the risk of hypersphere collapse inherent in the OCC objective, we introduce an inter-expert contrastive learning mechanism, with both theoretical proofs and empirical validation. As empirically demonstrated in Fig. 2, $S^2\text{MOE-F}$ establishes a more effective defense line against fake news than its single-expert counterparts (GIN [48] and Llama3 [8]). The main contributions of this paper are:

- Our novel MoE framework, $S^2\text{MOE-F}$, provides a robust defense against structural and semantic camouflage for fake news detection by allowing critical cross-validation between the distinct judgments of GNN and LLM experts.
- $S^2\text{MOE-F}$ has high practical value for realistic semi-supervised scenarios, breaking the dependency on expensive fake news labeling via the OCC objective while exploiting abundant unlabeled data through experts' diverse insights.
- We introduce a disentangled Transformer module to enforce expert specialization by reducing inter-expert redundancy, complemented by a novel RL-based router that dynamically determines the dominant expert for input data.
- Extensive experiments on diverse, real-world datasets sourced from prominent Web platforms and social media demonstrate the superior performance and practical applicability of $S^2\text{MOE-F}$ in detecting Web-spread fake news.

2 Preliminary and Related Work

This section outlines the preliminaries and briefly reviews related work. A more detailed literature review is available in Appendix A.

Semi-Supervised Fake News Detection. Let $\mathcal{D} = \mathcal{D}_l \cup \mathcal{D}_u$ denote the news dataset, where $\mathcal{D}_l = \{d_1, \dots, d_l\}$ is the subset of labeled news associated with labels $\{y_1, \dots, y_l\}$, as $y_i = 1$ indicates true news and $y_i = -1$ indicates fake news. The remaining subset $\mathcal{D}_u = \{d_{l+1}, \dots, d_N\}$ contains unlabeled news. The work will learn a fake news detector, denoted as $\mathcal{F}$, across the entire dataset $\mathcal{D} = \mathcal{D}_l \cup \mathcal{D}_u$. The learned detector is a function $\mathcal{F} : \mathcal{D} \rightarrow \mathbb{R}$ that maps each news item d_i to a misinformation score s_i , where a higher score s_i indicates a greater probability that the news is fake.

News Embedding by GNN. Each news item d_i is represented as a propagation graph $\mathcal{G}_i = (\mathcal{V}_i, \mathcal{E}_i)$, where the node set $\mathcal{V}_i$ comprises the source news item, as well as any interacting users and linked news articles, and the edge set $\mathcal{E}_i$ captures interactions (e.g., retweets, replies) between them. The GNN expert then encodes each graph $\mathcal{G}_i$ into a structural representation vector $\mathbf{h}_i \in \mathbb{R}^d$.

News Embedding by LLM. Each news article $d_i \in \mathcal{D}$ is encoded by LLMs to capture its semantic features. Specifically, we feed the textual content (e.g., title and content) of d_i into the LLM expert to obtain a contextualized semantic representation vector $\mathbf{z}_i \in \mathbb{R}^d$.

3 Method

3.1 Overall Architecture

The architecture of $S^2\text{MOE-F}$ is illustrated in Fig. 3. Firstly, GNN and LLM experts generate structural and semantic embeddings for each input news, respectively. These embeddings are then fed into a Disentanglement Mask Transformer to reduce inter-expert redundancy and integrate into a shared latent space. Then, $S^2\text{MOE-F}$ is trained with a comprehensive semi-supervised strategy: a One-Class Classification (OCC) objective constrains true news into a compact hypersphere, thereby identifying fake samples as out-of-boundary points; a contrastive loss ensures stability by preventing hypersphere collapse; and a pseudo-labeling scheme exploits unlabeled data via experts' consensus or divergence. Finally, a novel reinforcement learning (RL) router learns a dynamic policy to determine the dominant expert for input without explicit supervision.

3.2 Disentangled Masked Transformer

For each news item d_i , our GNN and LLM experts generate a structural embedding $\mathbf{h}_i \in \mathbb{R}^d$ and a semantic embedding $\mathbf{z}_i \in \mathbb{R}^d$, respectively (See Appendix B.1 for details). However, a significant redundancy arises between these two representations because they are derived from a shared textual foundation: the news content serves as the direct input for the LLM, while also being used to initialize the node features for the GNN's propagation graph. To mitigate this redundancy and enforce expert specialization, we introduce the fine-grained disentanglement of the two expert representations using a cross-expert attention mechanism. Firstly, we partition each embedding into K equal, non-overlapping pieces (or sub-embeddings), as:

$$\mathbf{h}_i = [\mathbf{h}_i^1, \mathbf{h}_i^2, \dots, \mathbf{h}_i^K], \quad \mathbf{z}_i = [\mathbf{z}_i^1, \mathbf{z}_i^2, \dots, \mathbf{z}_i^K], \quad (1)$$

where each piece $\mathbf{h}_i^k, \mathbf{z}_i^k \in \mathbb{R}^{d_k}$ with $d_k = d/K$. We then form matrix representations $\mathbf{H}_i, \mathbf{Z}_i \in \mathbb{R}^{d_k \times d_k}$ for use in the attention mechanism. Following the standard Transformer architecture [41], these are projected into Query (Q), Key (K), and Value (V) spaces using learnable weight matrices $\mathbf{W}_Q, \mathbf{W}_K, \mathbf{W}_V \in \mathbb{R}^{d_k \times d_k}$.

To follow up, we employ two asymmetric masked attention pathways. In this mechanism, a learnable soft mask matrix $\mathbf{M}$ is applied element-wise to the softmax-normalized attention scores. This allows the model to learn which cross-expert interactions to amplify or suppress before the final value aggregation.

In terms of the GNN-to-LLM pathway, the GNN representation queries the LLM representation to filter redundancy, and obtains the refined GNN sub-embedding matrix $\tilde{\mathbf{H}}_i$, as:

$$\tilde{\mathbf{H}}_i = \left(\mathbf{M}_1 \odot \text{softmax} \left(\frac{\mathbf{Q}_i \mathbf{K}_i^\top}{\sqrt{d_k}} \right) \right) \mathbf{V}_i, \quad (2)$$

where the matrices $\mathbf{Q}_i, \mathbf{K}_i, \mathbf{V}_i \in \mathbb{R}^{K \times d_k}$ are defined as $\mathbf{Q}_i = \mathbf{H}_i \mathbf{W}_Q, \mathbf{K}_i = \mathbf{H}_i \mathbf{W}_K, \mathbf{V}_i = \mathbf{Z}_i \mathbf{W}_V, \mathbf{M}_1 \in \mathbb{R}^{K \times K}$ is the learnable soft mask matrix, and $\odot$ denotes element-wise multiplication.

Towards the LLM-to-GNN pathway, the roles are inverted to refine the LLM sub-embedding matrix $\tilde{\mathbf{Z}}_i$:

$$\tilde{\mathbf{Z}}_i = \left(\mathbf{M}_2 \odot \text{softmax} \left(\frac{\mathbf{Q}_i \mathbf{K}_i^\top}{\sqrt{d_k}} \right) \right) \mathbf{V}_i, \quad (3)$$

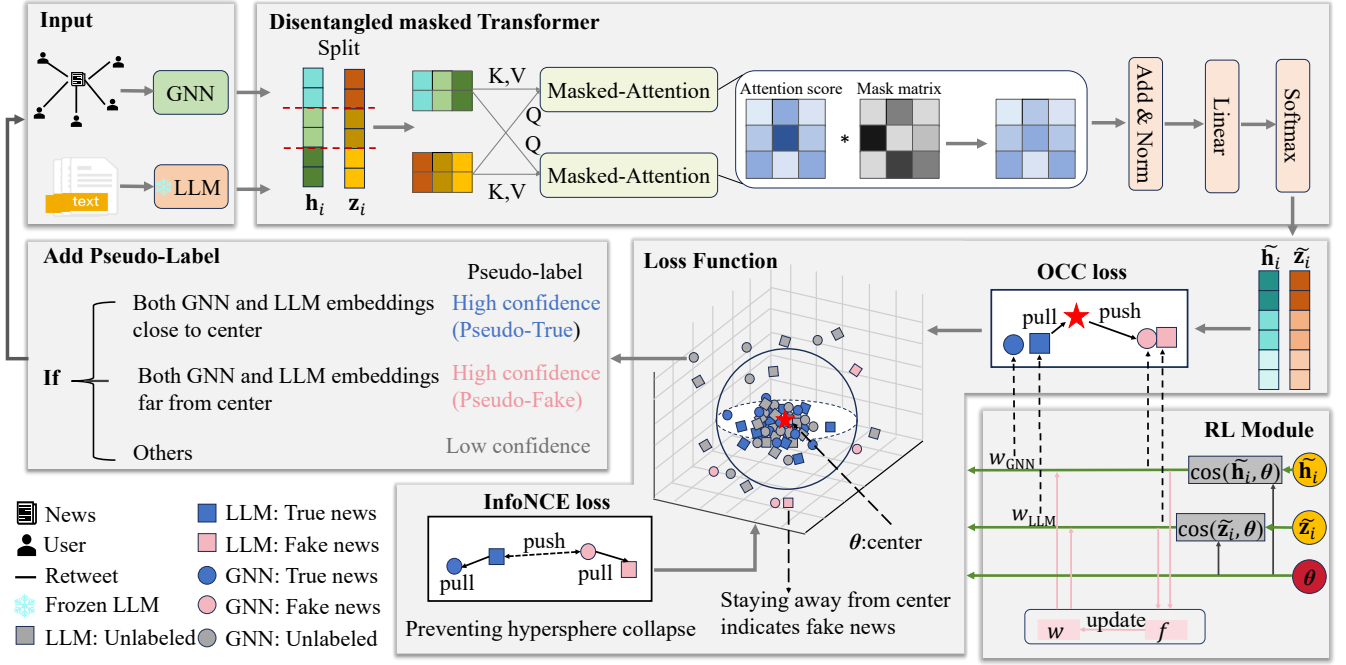


Figure 3: The architecture of S^2MOE-F . Taking the i -th news as an example, h_i and z_i are initial embeddings obtained by GNN and LLM experts, respectively. After redundancy removal, we obtain the refined embeddings $\tilde{h}_i, \tilde{z}_i$. We employ the cosine similarity $\cos(\cdot, \cdot)$ as the reward signal to serve the RL-policy function f that updates the weights w_{GNN} and w_{LLM} .

where $Q_i = Z_i W_Q, K_i = Z_i W_K, V_i = H_i W_V$, and $M_2 \in \mathbb{R}^{K \times K}$ is the second learnable soft mask matrix.

In both pathways, the attention outputs are matrix representations $\tilde{H}_i, \tilde{Z}_i \in \mathbb{R}^{K \times d_k}$, where each row corresponds to a refined sub-embedding. We concatenate these sub-embeddings to obtain the final refined embeddings $\tilde{h}_i \in \mathbb{R}^d$ and $\tilde{z}_i \in \mathbb{R}^d$ for the news d_i :

$$\tilde{h}_i = \text{Concat}[\tilde{h}_i^1, \tilde{h}_i^2, \dots, \tilde{h}_i^K], \quad \tilde{z}_i = \text{Concat}[\tilde{z}_i^1, \tilde{z}_i^2, \dots, \tilde{z}_i^K]. \quad (4)$$

To train these learnable soft mask matrices M_1 and M_2 to pursue meaningful disentanglement, we introduce a covariance-based optimization objective, $\mathcal{L}_{Cov}$. This loss penalizes the similarity between the covariance structures of the two final, refined embeddings:

$$\mathcal{L}_{Cov} = \frac{1}{(N-1)^2} \left\| (\tilde{H} - \tilde{H})^\top (\tilde{H} - \tilde{H}) - (\tilde{Z} - \tilde{Z})^\top (\tilde{Z} - \tilde{Z}) \right\|_F^2, \quad (5)$$

where $\tilde{H}, \tilde{Z} \in \mathbb{R}^{N \times d}$ are the refined embedding matrices for news, and $\|\cdot\|_F$ is the Frobenius norm. Minimizing this loss forces the masks to filter the attentions that reduce representational overlap, thereby encouraging each expert to preserve its specialization.

3.3 Training Objectives of the MoE

3.3.1 OCC loss. Refer to Section A, we identify One-Class Classification (OCC) as a particularly well-suited objective for realistic fake news detection and thus adopt it as the primary training objective for our MoE framework. This suitability stems from the inherent asymmetry in label acquisition for this task: large volumes of credible news can be readily collected from authoritative sources,

whereas annotating an item as "fake" is significantly more complex and costly, often requiring multi-party validation.

The goal of OCC is to learn a hypersphere, defined by a center θ , that encloses the representations of true news. Unlike prior work where this center is often fixed [34], we treat the center θ as a learnable parameter, allowing the model to find the optimal hypersphere center that covers true news. The OCC loss, $\mathcal{L}_{OCC}$, is formulated as a weighted sum of the squared Euclidean distances of each expert's embedding from this learnable center θ :

$$\mathcal{L}_{OCC} = w_{GNN} * \sum_{i=1}^N \|\tilde{h}_i - \theta\|_2^{y_i} + w_{LLM} * \sum_{i=1}^N \|\tilde{z}_i - \theta\|_2^{y_i}, \quad (6)$$

where $\tilde{h}_i$ and $\tilde{z}_i$ are the refined embeddings for news item d_i from the GNN and LLM experts, respectively. $\theta \in \mathbb{R}^d$ is the learnable center of the hypersphere, $y_i = 1$ indicates true news and $y_i = -1$ for fake news. The objective is to minimize the distance between true news and the center θ while maximizing it for fake news. And w_{GNN} and w_{LLM} are the dynamic weights for each expert, determined by the RL-based router in Section 3.5.

3.3.2 InfoNCE loss. However, a known issue with OCC objectives is the risk of *hypersphere collapse* [4, 32, 34], e.g., all output representations collapsing to a single point, a failure mode that we demonstrate theoretically in Section B.2. To counteract this, we incorporate a contrastive learning objective based on the Information

Noise Contrastive Estimation (InfoNCE) loss [35], defined as:

$$\mathcal{L}_{\text{InfoNCE}} = -\frac{1}{N} \sum_{i=1}^N \log \frac{\exp(\cos(\tilde{\mathbf{h}}_i, \tilde{\mathbf{z}}_i)/\tau)}{\sum_{k=1}^N \exp(\cos(\tilde{\mathbf{h}}_i, \tilde{\mathbf{z}}_k)/\tau)}, \quad (7)$$

where N is the number of news, $\cos(\cdot, \cdot)$ is the cosine similarity, and τ is a temperature hyperparameter. The principle of this loss is to pull the representations of the two expert views for the same news item ($\tilde{\mathbf{h}}_i$ and $\tilde{\mathbf{z}}_i$) closer together, while pushing them apart from the representations of other news items in the batch. As we also prove in Appendix B.2, the inclusion of the InfoNCE loss effectively prevents the *hypersphere collapse*.

3.3.3 Final loss. Finally, our composite training objective $\mathcal{L}$ is a weighted sum of the OCC loss ($\mathcal{L}_{\text{OCC}}$), the InfoNCE loss ($\mathcal{L}_{\text{InfoNCE}}$), and the covariance-based loss ($\mathcal{L}_{\text{Cov}}$), defined as:

$$\mathcal{L} = \lambda \mathcal{L}_{\text{OCC}} + \mu \mathcal{L}_{\text{InfoNCE}} + \gamma \mathcal{L}_{\text{Cov}}, \quad (8)$$

where λ , μ , and γ are non-negative hyperparameters that balance the contribution of each term. We present S²MOE-F's time complexity and pseudocode in Appendix B.3 and B.4, respectively.

3.4 Semi-Supervised Pseudo-Labeling Strategy

To leverage the abundant unlabeled data in $\mathcal{D}_u$, we introduce a strategy that assigns pseudo-labels to high-confidence samples. The confidence of a sample is determined by the consistency of the GNN and LLM experts' judgments relative to the learned OCC center θ . A high confidence score indicates that both experts agree on whether the sample is close to or far from the true news pattern. We formulate this confidence score, $\text{Conf}(\tilde{\mathbf{h}}_i, \tilde{\mathbf{z}}_i)$, for each unlabeled item $d_i \in \mathcal{D}_u$ as follows:

$$\text{Conf}(\tilde{\mathbf{h}}_i, \tilde{\mathbf{z}}_i) = 1 - \frac{||\tilde{\mathbf{h}}_i - \theta||_2 - ||\tilde{\mathbf{z}}_i - \theta||_2}{||\tilde{\mathbf{h}}_i - \theta||_2 + ||\tilde{\mathbf{z}}_i - \theta||_2 + \varepsilon}, \quad (9)$$

where θ is the center of the OCC hypersphere and ε is a small constant for numerical stability.

A high confidence score signifies strong consensus between the experts. For top- K high-confidence samples, we then assign relative pseudo-labels to them:

- **Pseudo-True** ($y_i = 1$): Assigned to high-confidence samples where both embeddings are close to the center θ .
- **Pseudo-Fake** ($y_i = -1$): Assigned to high-confidence samples where both embeddings are far from the center θ .
- **Low-Confidence**: Samples with a low confidence score, indicating experts' divergence on their distance to the center, are not assigned a pseudo-label.

Finally, these newly high-confidence pseudo-labels are added to the labeled set $\mathcal{D}_l$ for subsequent rounds of iterative training.

3.5 RL-based Routing Mechanism

As defined in our OCC objective in Eq. (6), the weights w_{GNN} and w_{LLM} modulate the respective dominance of the GNN and LLM experts. A static weighting scheme is suboptimal, as different samples may require different dominant experts. However, learning a per-sample routing policy is also unreliable in our semi-supervised setting. This is because the routing policy is guided by the model's

outputs rather than by ground-truth labels. Consequently, an erroneous judgment from an expert on a single sample can mislead the routing mechanism, causing the wrong expert to be chosen as dominant and propagating this error into subsequent training rounds. Instead, we propose a more robust, batch-aware routing strategy by introducing an RL-based router, which performs an exploration-exploitation process across the true news samples within each batch. This allows the router to converge on a set of stable weights, $\{w_{\text{GNN}}, w_{\text{LLM}}\}$, representing the learned dominance of the GNN and LLM experts for the current batch.

3.5.1 Reward Formulation. The reward signal, r , is derived from the labeled true news samples. We define r as the cosine similarity between an expert's embedding and the OCC center θ , as this provides a stable and well-scaled reward signal for the RL agent. For each true news item d_t , the per-sample rewards are:

$$r_{\text{GNN}}^t = \cos(\tilde{\mathbf{h}}_t, \theta), \quad r_{\text{LLM}}^t = \cos(\tilde{\mathbf{z}}_t, \theta). \quad (10)$$

3.5.2 Policy Update and Normalization. The weights for a given batch are determined via an iterative update process, initialized as $w_{\text{GNN}}^0 = w_{\text{LLM}}^0 = 0.5$. Let T be the number of labeled true news samples in the batch. Then, for each true news sample $t = 1, \dots, T$, we use its corresponding rewards ($r_{\text{GNN}}^t, r_{\text{LLM}}^t$) to update the weights:

$$\begin{aligned} w_{\text{GNN}}^{t'} &= w_{\text{GNN}}^{t-1'} + \eta \cdot (r_{\text{GNN}}^t - r_{\text{LLM}}^t), \\ w_{\text{LLM}}^{t'} &= w_{\text{LLM}}^{t-1'} - \eta \cdot (r_{\text{GNN}}^t - r_{\text{LLM}}^t), \end{aligned} \quad (11)$$

where η is a learning rate. After iterating through all T true news samples, we obtain $w_{\text{GNN}}^{T'}$ and $w_{\text{LLM}}^{T'}$. To ensure the updated weights remain a valid probability distribution, a normalization is applied,

$$w_{\text{GNN}} = \frac{\max(0, w_{\text{GNN}}^{T'})}{\max(0, w_{\text{GNN}}^{T'}) + \max(0, w_{\text{LLM}}^{T'}) + \varepsilon}. \quad (12)$$

The final weight w_{LLM} is calculated analogously. These final, batch-aware weights are then used to compute the OCC loss in Eq. (6), for all samples in the current batch.

3.6 Inference Stage

Once the S²MOE-F is trained, it can be used to measure a misinformation score s_i for an unseen news item d_i in the inference stage. For each news item d_i in the test set, we give its score:

$$s_i = w_{\text{GNN}} * ||\tilde{\mathbf{h}}_i - \theta||_2 + w_{\text{LLM}} * ||\tilde{\mathbf{z}}_i - \theta||_2. \quad (13)$$

where a higher score s_i indicates a greater probability that the news is fake. However, the weights w_{GNN} and w_{LLM} cannot be computed directly on the test set, as their calculation by RL router (Section 3.5) requires labeled true news samples, which are unavailable during inference. To address this, we incorporate the pseudo-labeling strategy (Section 3.4) with the RL-router (Section 3.5). Specifically, we pseudo-label a subset of reliable "pseudo-true" examples with high confidence in a test batch. These "pseudo-true" examples can be used for the RL-based routing mechanism to dynamically adjust w_{GNN} and w_{LLM} during inference.

Table 1: Results (%) of S^2MOE-F^L and S^2MOE-F^Q , compared with baseline methods on the five datasets. Gray highlighting denotes the best performance, while underlining indicates the runner-up.

Methods	COVID-19		GossipCop		PolitiFact		Weibo-21		CTI	
	AUC $\uparrow$	Pre $\uparrow$	AUC $\uparrow$	Pre $\uparrow$	AUC $\uparrow$	Pre $\uparrow$	AUC $\uparrow$	Pre $\uparrow$	AUC $\uparrow$	Pre $\uparrow$
BERT	64.73 $\pm$ 3.58	65.18 $\pm$ 3.09	73.52 $\pm$ 4.27	66.82 $\pm$ 3.64	64.21 $\pm$ 2.98	72.51 $\pm$ 3.64	71.94 $\pm$ 3.92	67.58 $\pm$ 5.22	70.60 $\pm$ 1.97	<u>70.41$\pm$3.41</u>
XLNet	68.42 $\pm$ 6.81	75.52 $\pm$ 7.15	77.79 $\pm$ 6.44	73.62 $\pm$ 2.69	78.26 $\pm$ 1.20	78.33 $\pm$ 1.95	72.13 $\pm$ 4.23	67.80 $\pm$ 6.38	71.75 $\pm$ 3.52	67.69 $\pm$ 4.98
OCC	72.86 $\pm$ 6.51	75.82 $\pm$ 3.81	78.47 $\pm$ 6.38	71.97 $\pm$ 5.75	69.02 $\pm$ 8.53	74.06 $\pm$ 2.74	68.28 $\pm$ 4.18	68.37 $\pm$ 3.57	70.56 $\pm$ 5.71	69.74 $\pm$ 4.47
UFD	76.38 $\pm$ 5.28	68.47 $\pm$ 5.82	72.25 $\pm$ 4.73	66.07 $\pm$ 3.68	69.36 $\pm$ 2.13	70.62 $\pm$ 4.12	67.32 $\pm$ 3.71	66.29 $\pm$ 3.27	65.03 $\pm$ 6.19	68.29 $\pm$ 6.16
GCN-FN	73.22 $\pm$ 2.99	69.31 $\pm$ 1.55	77.00 $\pm$ 5.71	72.72 $\pm$ 8.06	76.53 $\pm$ 2.14	76.88 $\pm$ 3.71	70.14 $\pm$ 0.58	64.82 $\pm$ 2.75	69.99 $\pm$ 1.50	65.58 $\pm$ 1.39
GNN-CL	78.76 $\pm$ 1.43	73.42 $\pm$ 1.82	77.57 $\pm$ 3.58	75.17 $\pm$ 6.61	77.03 $\pm$ 2.15	74.28 $\pm$ 3.51	70.19 $\pm$ 0.96	64.82 $\pm$ 1.70	67.16 $\pm$ 2.53	63.18 $\pm$ 2.02
GNN	74.21 $\pm$ 3.21	71.34 $\pm$ 4.57	77.58 $\pm$ 0.37	67.12 $\pm$ 0.43	73.28 $\pm$ 0.35	61.58 $\pm$ 0.36	69.83 $\pm$ 0.43	74.11 $\pm$ 1.14	66.62 $\pm$ 2.39	55.77 $\pm$ 3.05
ARG	74.44 $\pm$ 4.98	68.83 $\pm$ 2.60	73.54 $\pm$ 1.59	71.81 $\pm$ 0.65	75.97 $\pm$ 0.47	79.12 $\pm$ 0.61	62.66 $\pm$ 3.81	69.56 $\pm$ 1.63	63.22 $\pm$ 4.98	68.83 $\pm$ 2.60
GAMC	70.51 $\pm$ 3.59	69.41 $\pm$ 7.43	62.89 $\pm$ 3.68	74.42 $\pm$ 3.68	65.70 $\pm$ 8.92	58.89 $\pm$ 4.88	62.83 $\pm$ 3.71	60.41 $\pm$ 4.17	63.81 $\pm$ 6.21	69.37 $\pm$ 5.18
InfOp	78.16 $\pm$ 3.81	73.59 $\pm$ 4.68	75.82 $\pm$ 5.45	72.01 $\pm$ 5.64	74.32 $\pm$ 4.44	71.05 $\pm$ 2.34	67.46 $\pm$ 3.56	68.92 $\pm$ 6.48	65.32 $\pm$ 1.46	67.43 $\pm$ 5.01
MFCL	75.10 $\pm$ 3.87	69.15 $\pm$ 3.34	74.66 $\pm$ 3.24	76.27 $\pm$ 5.89	66.91 $\pm$ 5.14	70.85 $\pm$ 2.52	65.58 $\pm$ 1.41	60.37 $\pm$ 5.18	63.40 $\pm$ 0.46	65.32 $\pm$ 1.93
S^2MOE-F^Q	81.04 $\pm$ 3.25	77.35 $\pm$ 2.53	78.83 $\pm$ 2.23	76.11 $\pm$ 3.02	78.55 $\pm$ 1.59	79.44 $\pm$ 1.22	76.16 $\pm$ 1.93	76.12 $\pm$ 3.25	72.03 $\pm$ 1.93	67.28 $\pm$ 2.34
S^2MOE-F^L	84.21 $\pm$ 3.35	81.40 $\pm$ 2.12	80.13 $\pm$ 2.06	79.23 $\pm$ 2.81	80.03 $\pm$ 1.59	79.76 $\pm$ 1.24	76.85 $\pm$ 2.17	78.37 $\pm$ 1.40	73.68 $\pm$ 2.23	71.65 $\pm$ 2.37

Methods	COVID-19		GossipCop		PolitiFact		Weibo-21		CTI	
	Rec $\uparrow$	F1 $\uparrow$	Rec $\uparrow$	F1 $\uparrow$	Rec $\uparrow$	F1 $\uparrow$	Rec $\uparrow$	F1 $\uparrow$	Rec $\uparrow$	F1 $\uparrow$
BERT	67.35 $\pm$ 2.97	74.16 $\pm$ 4.51	59.42 $\pm$ 2.61	62.53 $\pm$ 2.64	71.43 $\pm$ 2.88	75.68 $\pm$ 3.28	67.81 $\pm$ 2.78	67.30 $\pm$ 6.31	69.64 $\pm$ 3.45	<u>72.11$\pm$3.89</u>
XLNet	57.68 $\pm$ 3.82	58.31 $\pm$ 5.81	63.28 $\pm$ 5.37	67.34 $\pm$ 4.31	74.03 $\pm$ 5.33	76.59 $\pm$ 3.75	67.62 $\pm$ 3.17	67.38 $\pm$ 6.18	65.08 $\pm$ 5.82	68.26 $\pm$ 2.64
OCC	65.82 $\pm$ 5.81	71.89 $\pm$ 5.62	72.40 $\pm$ 4.64	78.66 $\pm$ 3.63	76.42 $\pm$ 3.83	61.06 $\pm$ 4.03	68.31 $\pm$ 3.19	65.17 $\pm$ 1.49	62.91 $\pm$ 3.29	65.13 $\pm$ 3.82
UFD	68.37 $\pm$ 4.19	70.62 $\pm$ 5.29	64.73 $\pm$ 4.45	67.16 $\pm$ 4.31	65.27 $\pm$ 1.36	67.12 $\pm$ 2.74	62.17 $\pm$ 2.73	68.25 $\pm$ 2.44	68.32 $\pm$ 4.13	61.05 $\pm$ 4.16
GCN-FN	72.50 $\pm$ 4.08	71.38 $\pm$ 6.52	71.01 $\pm$ 4.50	76.99 $\pm$ 5.61	74.62 $\pm$ 2.54	76.35 $\pm$ 2.47	<u>72.42$\pm$3.75</u>	69.06 $\pm$ 1.87	63.93 $\pm$ 2.30	64.36 $\pm$ 1.03
GNN-CL	63.88 $\pm$ 4.54	69.67 $\pm$ 1.83	70.57 $\pm$ 4.55	68.25 $\pm$ 6.64	76.18 $\pm$ 3.52	75.28 $\pm$ 3.52	71.15 $\pm$ 2.44	63.63 $\pm$ 1.39	68.59 $\pm$ 0.23	64.39 $\pm$ 2.64
GNN	66.14 $\pm$ 2.78	70.46 $\pm$ 1.91	75.18 $\pm$ 0.58	70.98 $\pm$ 4.41	77.16 $\pm$ 0.60	68.25 $\pm$ 1.39	70.88 $\pm$ 1.66	62.59 $\pm$ 0.98	68.78 $\pm$ 4.98	58.96 $\pm$ 3.44
ARG	68.80 $\pm$ 2.58	68.35 $\pm$ 2.38	75.45 $\pm$ 0.37	70.08 $\pm$ 0.17	62.97 $\pm$ 0.18	61.08 $\pm$ 0.17	68.49 $\pm$ 1.96	67.77 $\pm$ 1.84	68.80 $\pm$ 2.58	68.74 $\pm$ 3.19
GAMC	73.92 $\pm$ 7.29	68.41 $\pm$ 4.91	67.86 $\pm$ 1.00	58.30 $\pm$ 3.36	76.35 $\pm$ 0.30	66.97 $\pm$ 5.36	64.29 $\pm$ 1.73	65.73 $\pm$ 7.42	61.37 $\pm$ 3.91	68.29 $\pm$ 7.15
InfOp	71.54 $\pm$ 3.12	68.52 $\pm$ 2.73	74.51 $\pm$ 3.49	64.14 $\pm$ 3.34	70.32 $\pm$ 3.46	61.24 $\pm$ 5.33	66.02 $\pm$ 5.49	66.79 $\pm$ 4.41	64.37 $\pm$ 4.99	67.29 $\pm$ 3.72
MFCL	70.32 $\pm$ 3.46	64.21 $\pm$ 1.33	71.43 $\pm$ 3.64	65.30 $\pm$ 2.38	67.43 $\pm$ 4.56	67.44 $\pm$ 5.28	63.84 $\pm$ 4.95	63.29 $\pm$ 5.27	63.63 $\pm$ 4.31	64.98 $\pm$ 4.24
S^2MOE-F^Q	<u>74.63$\pm$3.25</u>	<u>74.35$\pm$3.16</u>	<u>76.15$\pm$2.69</u>	<u>79.36$\pm$3.07</u>	<u>77.93$\pm$2.98</u>	<u>77.10$\pm$3.07</u>	68.42 $\pm$ 2.54	<u>74.25$\pm$2.74</u>	<u>69.67$\pm$3.93</u>	69.91 $\pm$ 2.53
S^2MOE-F^L	78.41 $\pm$ 3.54	78.79 $\pm$ 3.22	78.45 $\pm$ 1.05	79.41 $\pm$ 2.14	78.93 $\pm$ 0.85	77.97 $\pm$ 1.37	72.97 $\pm$ 2.13	74.83 $\pm$ 2.39	72.61 $\pm$ 2.98	76.59 $\pm$ 3.58

4 Experiment

In this section, we conduct extensive experiments on five datasets to evaluate S^2MOE-F , including ablation studies for module effectiveness, case studies to validate redundancy removal by the disentangled masked transformer, parameter sensitivity analysis, and evaluations of running time and memory usage.

4.1 Experimental Setup

4.1.1 Datasets. GossipCop, PolitiFact, COVID-19, and Weibo-21, are four real-world fake news datasets, and CTI is a news dataset generated by LLM. GossipCop and PolitiFact both come from FakeNewsNet [38]. GossipCop mainly covers entertainment gossip, while PolitiFact covers statements or reports on political topics in the US. COVID-19 [50] includes news reports and health statements related to the pandemic, and Weibo-21 [28] includes news, user posts, and interactions collected from the Weibo platform, CTI [5] includes various types of news generated by LLM.

4.1.2 Baseline & Experimental Settings. To demonstrate the effectiveness of S^2MOE-F , we compare it with several baselines, including: BERT [7], XLNet [54], OCC [34], UFD [53], GCN-FN [26], GNN-CL [11], GNN [49], GAMC [59], ARG [12], InfOp [47], and

MFCL [6]. To follow a standard semi-supervised setting [10, 20, 27], we utilize only 30% of the data with labels, while the remaining training data and the entire test set are treated as unlabeled. We report the average results and standard deviations across the four most commonly used metrics under 5-fold cross-validation. Consequently, both S^2MOE-F and all baselines are trained in the same semi-supervised setting. More details of the experimental setting, including baseline summary and parameter settings, are shown in the Appendix C.1.

In our experiments, we further evaluate S^2MOE-F with two different LLMs as the semantic backbone: Llama3 and Qwen. The version using Llama3 is denoted as S^2MOE-F^L , while the one using Qwen is denoted as S^2MOE-F^Q . Among them, S^2MOE-F^L serves as our primary model, as it achieves superior performance across all evaluation metrics.

4.2 Detection Result

Table 1 reports the performance comparison between S^2MOE-F and a series of baseline methods across five benchmark datasets (COVID-19, GossipCop, PolitiFact, Weibo-21, and CTI). The evaluation is conducted using four commonly adopted metrics: AUC, Precision, Recall, and F1-score. As shown in the Table1, both S^2MOE-F^L and

Table 2: Ablation studies of S^2MOE-F^L on the five datasets. Gray highlighting denotes the best performance.

Protocols	COVID-19		GossipCop		PolitiFact		Weibo-21		CTI	
	AUC $\uparrow$	F1 $\uparrow$	AUC $\uparrow$	F1 $\uparrow$	AUC $\uparrow$	F1 $\uparrow$	AUC $\uparrow$	F1 $\uparrow$	AUC $\uparrow$	F1 $\uparrow$
$S^2MOE-F^L-\mathcal{O}GNN$	74.02 $\pm$ 2.05	71.39 $\pm$ 3.16	70.39 $\pm$ 3.58	69.72 $\pm$ 3.04	71.94 $\pm$ 5.12	68.61 $\pm$ 2.46	70.11 $\pm$ 1.57	63.38 $\pm$ 4.09	66.42 $\pm$ 2.58	63.08 $\pm$ 6.32
$S^2MOE-F^L-\mathcal{O}Llama$	71.39 $\pm$ 2.71	69.03 $\pm$ 3.81	71.24 $\pm$ 2.45	70.32 $\pm$ 3.14	68.37 $\pm$ 3.36	71.82 $\pm$ 2.76	69.75 $\pm$ 2.84	63.23 $\pm$ 4.70	64.05 $\pm$ 3.43	62.01 $\pm$ 4.76
$S^2MOE-F^L-\mathcal{O}RL$	81.03 $\pm$ 3.68	75.90 $\pm$ 4.41	78.54 $\pm$ 3.05	75.02 $\pm$ 4.62	78.55 $\pm$ 0.80	75.49 $\pm$ 2.59	75.12 $\pm$ 1.65	68.68 $\pm$ 1.46	71.53 $\pm$ 3.36	69.56 $\pm$ 5.76
$S^2MOE-F^L-\mathcal{O}L_{OCC}$	71.64 $\pm$ 2.84	65.48 $\pm$ 3.05	67.54 $\pm$ 4.13	67.02 $\pm$ 5.66	67.71 $\pm$ 1.76	67.92 $\pm$ 2.28	66.06 $\pm$ 1.86	66.75 $\pm$ 1.60	63.89 $\pm$ 4.05	65.31 $\pm$ 3.80
$S^2MOE-F^L-\mathcal{O}L_{Cov}$	77.30 $\pm$ 3.68	74.41 $\pm$ 4.76	76.45 $\pm$ 3.33	74.45 $\pm$ 1.15	77.06 $\pm$ 1.43	74.51 $\pm$ 1.51	74.49 $\pm$ 4.28	65.01 $\pm$ 2.61	70.23 $\pm$ 1.49	69.68 $\pm$ 7.11
$S^2MOE-F^L-\mathcal{O}L_{InfoNCE}$	73.17 $\pm$ 2.49	67.35 $\pm$ 2.41	68.15 $\pm$ 1.37	68.33 $\pm$ 2.04	69.48 $\pm$ 1.94	67.40 $\pm$ 2.49	68.75 $\pm$ 1.25	65.24 $\pm$ 1.05	63.03 $\pm$ 1.43	68.13 $\pm$ 7.30
S^2MOE-F^L	84.21 $\pm$ 3.35	78.79 $\pm$ 3.22	80.13 $\pm$ 2.06	79.41 $\pm$ 2.14	80.03 $\pm$ 1.59	77.97 $\pm$ 1.37	76.85 $\pm$ 2.17	74.83 $\pm$ 2.39	73.68 $\pm$ 2.23	76.59 $\pm$ 3.58

S^2MOE-F^Q consistently outperform all baselines across almost all datasets and metrics. Compared with recent baselines, including GAMC, InfOp, and MFCL, S^2MOE-F^L achieves notable gains in AUC (by approximately 3–6%) and F1-score (up to 10%), demonstrating the effectiveness of integrating semantic and structural information within our MoE framework. Furthermore, S^2MOE-F^L achieves higher performance than S^2MOE-F^Q on all datasets, indicating that the stronger semantic modeling capability of Llama3 contributes to more accurate and robust fake news detection. These results collectively verify the superiority and robustness of S^2MOE-F across diverse datasets.

4.3 Ablation Study

In this section, we demonstrate the necessity of each module in S^2MOE-F by conducting ablation studies. We use S^2MOE-F^L in ablation studies due to its better performance, where we observe the model’s performance after removing each module to assess their impact on S^2MOE-F^L :

1) $S^2MOE-F^L-\mathcal{O}GNN$ removes the GNN and trains the model only on text data; 2) $S^2MOE-F^L-\mathcal{O}Llama$ removes the LLM and trains the model only on graph-structured data; 3) $S^2MOE-F^L-\mathcal{O}RL$ removes the RL-based router, when calculating OCC loss, fixing the weights of GNN and LLM; 4) $S^2MOE-F^L-\mathcal{O}L_{OCC}$ removes the OCC loss and only depends on the contrast and covariance losses to optimize the model; 5) $S^2MOE-F^L-\mathcal{O}L_{Cov}$ removes the covariance loss and uses two fixed mask matrices when disentangling; 6) $S^2MOE-F^L-\mathcal{O}L_{InfoNCE}$ removes contrastive loss and depends on OCC and covariance losses to evaluate model performance.

Table 2 demonstrates that the complete S^2MOE-F^L achieves the best overall performance across all datasets, highlighting the necessity of each component. $S^2MOE-F^L-\mathcal{O}GNN$ shows a significant drop in performance, indicating that removing the GNN leads to a loss of propagation structure information, which is critical for misinformation detection. $S^2MOE-F^L-\mathcal{O}Llama$ also causes a notable decline, highlighting the importance of semantic understanding provided by the LLM. When the RL-based router is removed, performance consistently decreases, showing that dynamically balancing the contributions of GNN and LLM is beneficial for improving performance. Without OCC loss, depending only on assessing similarities and differences between news is insufficient to effectively distinguish true and fake news, while excluding the covariance loss degrades performance on datasets, confirming its role in disentangling redundant features. Moreover, removing contrastive learning loss increases the risk of *hypersphere collapse*, further emphasizing

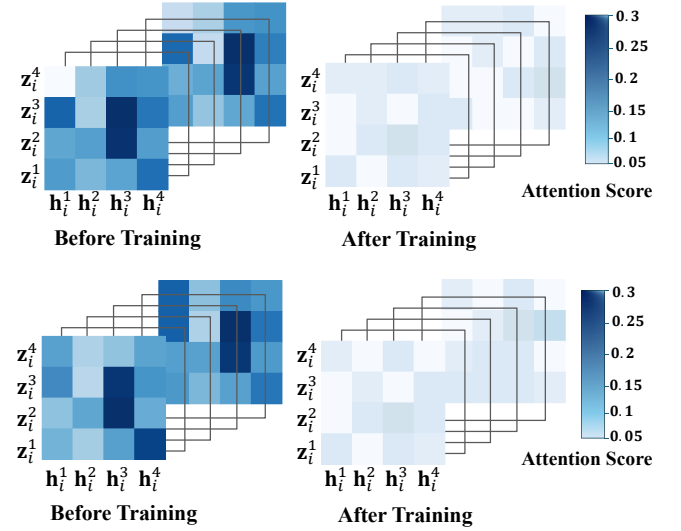


Figure 4: Attention score matrices between the GNN ($h_i^1 \sim h_i^4$) and LLM ($z_i^1 \sim z_i^4$) sub-embeddings before and after Disentangled Masked Transformer training. Take two data from a batch as an example. The left image shows dense cross-expert attention that indicates redundancy, while the right image becomes more diagonal and sparse, showing that the Disentangled Masked Transformer alleviates redundancy and enhances expert specialization.

its necessity. These results clearly validate that the integration of all components is essential for the robustness of S^2MOE-F^L .

4.4 Case Study of Disentangled Masked Transformer

Fig 4 illustrates the redundancy removal process of the disentanglement module, comparing the interactions between sub-components of GNN embeddings (h_i) and LLM embeddings (z_i) before and after training. The color bar on the right quantifies the magnitude of interaction intensity between sub-components — darker shades indicate stronger and more redundant interactions, while lighter shades represent weaker or less redundant ones. Before training, the heatmaps exhibit numerous dark-colored regions, indicating significant and extensive redundant correlations across different

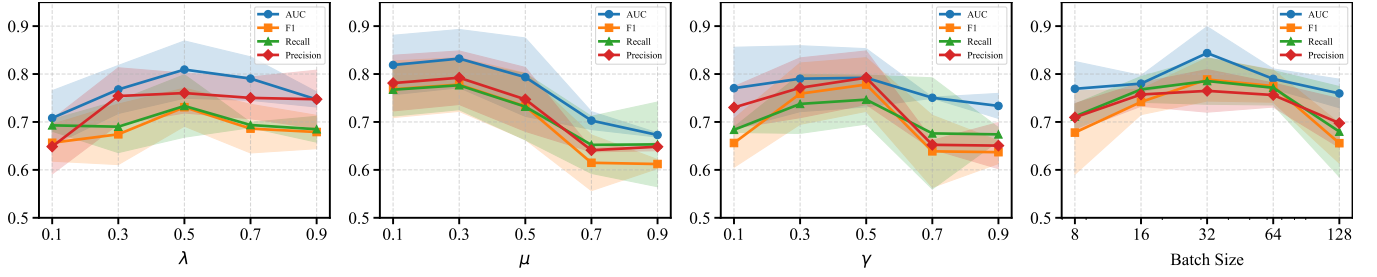


Figure 5: Parameter analysis for λ , μ , γ and batch size in S^2 MOE-F on the COVID-19 dataset.

sub-components. After training, these dark regions are drastically reduced, with the overall color distribution becoming lighter and sparser, aligning with the color bar's indication of diminished interaction intensity. This phenomenon directly demonstrates that the disentanglement module, through the cross-expert attention mechanism and learnable masks, effectively filters redundant interactions. As a result, the sub-components of GNN and LLM embeddings can focus more on their specialized representations, validating the module's efficacy in reducing redundancy and enhancing the specificity of expert representations.

4.5 Parameter Analysis

In this section, we analyze the parameters to ensure the best performance of S^2 MOE-F. Fig. 5 presents the parameter sensitivity analysis of λ , μ , γ , and batch size on the COVID-19 dataset, where λ , μ , and γ correspond to the weights of the OCC loss ($\mathcal{L}_{OCC}$), contrastive loss ($\mathcal{L}_{InfoNCE}$), and covariance loss ($\mathcal{L}_{Cov}$) in Eq. (8), respectively. As shown in the figure, the model achieves the best overall performance when λ , μ , and γ are set around 0.5. Specifically, as λ increases from 0.1 to 0.5, all evaluation metrics (AUC, F1, Recall, and Precision) consistently improve, indicating that a moderate weighting of $\mathcal{L}_{OCC}$ effectively enhances the model's capability to distinguish fake news. However, further increasing λ beyond 0.5 leads to a slight performance decline, as overemphasizing $\mathcal{L}_{OCC}$ reduces the contribution of auxiliary objectives. For μ , which controls the contrastive loss $\mathcal{L}_{InfoNCE}$, the performance gradually decreases as μ grows, suggesting that a small coefficient is sufficient to prevent *hypersphere collapse*. Similarly, for γ , which regulates the covariance loss $\mathcal{L}_{Cov}$, the model performs best when γ is within the range of 0.3–0.5. This implies that $\mathcal{L}_{Cov}$ effectively assists the disentanglement module by guiding the learning of two mask matrices, while excessive weighting reduces feature separability. Finally, we analyze the impact of batch size. As shown in the last subplot of Fig. 5, increasing the batch size generally improves the performance up to 32, where all metrics reach their peak. Batch sizes larger than 32 lead to a slight decrease in some metrics, which suggests that too large a batch size may reduce the generalization ability of the model.

4.6 Time and Memory Efficiency Analysis

We analyze the efficiency of different models on the COVID-19 dataset in terms of training time and GPU memory usage, as illustrated in Fig. 6. The proposed S^2 MOE-F achieves the highest AUC (≈ 0.85) with moderate computational cost (≈ 250 ms per step) and

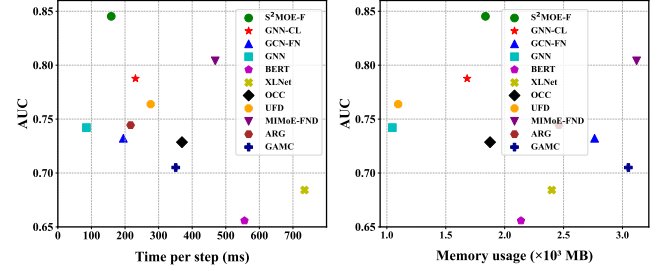


Figure 6: Efficiency analysis on COVID-19 dataset.

relatively low memory usage ($\approx 1.5 \times 10^3$ MB), demonstrating superior scalability and efficiency in practical settings. In comparison, GNN-CL attains slightly lower AUC but faster convergence, showing a favorable trade-off between accuracy and efficiency across different metrics. Transformer-based baselines such as BERT and XLNet exhibit much higher time and memory consumption (over 600 ms per step and 2×10^3 MB) yet fail to achieve competitive performance, highlighting their limited efficiency in this task. Simpler GNN variants (e.g., GCN-FN, GNN-CL, GNN) are faster but less accurate. Overall, S^2 MOE-F provides the best balance between performance, training speed, and memory footprint, confirming the advantage of MoE.

5 Conclusion

In this paper, we propose S^2 MOE-F, a semi-supervised mixture-of-experts (MoE) framework for fake news detection that leverages the complementary strengths of the Graph Neural Network (GNN) and Large Language Model (LLM) experts. Our approach mitigates inter-expert redundancy with a disentangled masked Transformer, employs an One-Class Classification (OCC) objective and pseudo-labeling to address label scarcity, and introduces an RL-based router for dynamic expert weighting. Extensive experiments on real-world datasets demonstrate that S^2 MOE-F achieves superior performance and provides a robust defense against both semantic and structural camouflage, even with few labels. These results highlight a promising direction that uses MoE frameworks for realistic semi-supervised fake news detection.

Acknowledgments

This work is supported by the National Natural Science Foundation of China (NSFC, Grant No. 52227901, 62405006, 62501014).

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A Related Work

Context-based Fake News Detection. There are some mature methods for fake news detection based on text content, for example, a recent work proposed a deep multiple instance learning framework, Conv-ADMIL [14], which integrates attention mechanisms with contextual neural language models, which enhanced the model’s ability to distinguish between true and false news. Some other methods detect fake news in literature by capturing the style characteristics of fake news [31] and factual reasoning [37, 45]. Wu et al. [46] proposed a multimodal fusion method that used a collaborative attention network to integrate text and visual information to improve the detection accuracy. Dun et al. [9] developed KAN, which incorporated external knowledge into the attention mechanism to improve the model’s understanding of context. Singhal et al. [39] introduced SpotFake+, which combined textual and visual information through transfer learning to provide more powerful detection results. These studies highlight the effectiveness of content-based approaches for detecting fake news, demonstrating their ability to capture semantic cues indicative of misinformation.

Graph-based Fake News Detection. In recent years, using graph structures to detect fake news has attracted wider attention [57]. The ability of GNN to model complex relations demonstrates promising progress in this field. GCAN [21] achieves interpretable fake news detection by combining the graph structure and the collaborative attention mechanism. Bian et al. [2] used a bi-directional graph convolutional network to effectively capture the complexity of information interaction on the social network for fake news detection. Nguyen et al. [29] designed the FANG model, which used graph representation learning methods to strengthen the support of the social context for fake news detection. Min et al. [24] developed the Post-User Interaction Network, which adopted a divide-and-conquer strategy to accurately model the interactive relationship

between users and posts. In addition, Phan et al. [30] conducted a comprehensive review of fake news detection methods based on graph neural networks, summarizing the advantages and application scenarios of current technologies. These studies demonstrate that graph-based approaches not only improve the accuracy of fake news detection systems but also enhance their interpretability. However, current methods still have many drawbacks. Most of them rely on supervision and a single view.

One Class Classification. One-Class Classification (OCC) is a powerful technique for detecting anomalies, particularly useful when only true news or very limited fake news is available. Ruff et al. [34] introduced Deep One-Class Classification, using deep neural networks to map data into a compact latent space, enhancing generalization. Chalapathy et al. [4] combined autoencoders with one-class Support Vector Machine (SVM) to detect anomalies in high-dimensional data. Qiu et al. [32] extended this to graph-level anomaly detection [55, 58, 60–62] by integrating GNNs with OCC techniques, effectively capturing structural anomalies in graphs. These advancements demonstrated the potential of deep learning and graph-based methods in handling complex and high-dimensional data, providing robust solutions for various anomaly detection tasks. OCC methods typically aim to learn a decision boundary that separates normal data from potential outliers by enclosing the normal data points in a compact hypersphere space.

B Supplementary of Method

B.1 Feature Extraction: GNN and LLM

This section describes the process of extracting features from LLM [3, 7, 17, 33] and GNN [40, 48, 56], followed by a detailed explanation of feature extraction from the two experts.

For graph data, the GNN feature extractor extracts features, where each layer updates the node representation based on its neighbors. The GNN expert adopted in this paper is GIN [48]. Let $\mathbf{h}_v^{(l)}$ denote the updated representation vector of node v in layer l , $\mathbf{h}_v^{(l-1)}$ denote the vector of node v in previous layer $l-1$, $\mathbf{h}_u^{(l-1)}$ denote the vector of neighbor u of node v in the previous layer $l-1$, $\mathcal{V}_i(v)$ denotes the set of neighbors of node v .

$$\mathbf{h}_v^{(l)} = \text{GNN}^{(l)} \left(\mathbf{h}_v^{(l-1)}, \mathbf{h}_u^{(l-1)} \mid u \in \mathcal{V}_i(v) \right). \quad (14)$$

The node representations from each layer are combined to obtain the graph-level representation.

$$\mathbf{h}_{\mathcal{G}_i}^{(l)} = \text{READOUT}^{(l)} \left(\mathbf{h}_v^{(l)} \mid v \in \mathcal{G}_i \right). \quad (15)$$

Then combine all the graph-level representations of each layer. The vector of representations in the graph is denoted as $\mathbf{h}_i$.

For the textual content of the news, we use the LLM expert to extract features. The LLM expert used in this paper is Llama3 [8]. We only select the pre-trained Llama3 without fine-tuning, mainly aiming to reduce training costs. At the same time, we believe that with its large-scale pre-training and strong in-context learning capabilities, Llama3 is already well-suited for news semantic understanding. The text is preprocessed and converted into an input sequence $\mathbf{X} = (\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n)$ and the corresponding attention mask is also generated. The corresponding attention mask is also generated. The sequence is then inputted into the LLM to obtain

the contextualized token representation matrix:

$$\mathbf{Z}^{\text{text}} = \text{LLM}(X) = (\mathbf{z}_1^{\text{text}}, \mathbf{z}_2^{\text{text}}, \dots, \mathbf{z}_n^{\text{text}}), \quad \mathbf{z}_j^{\text{text}} \in \mathbb{R}^d. \quad (16)$$

Finally, after a pooling operation, we obtain the text-level representation of the i -th news: $\mathbf{z}_i = \text{POOL}(\mathbf{Z}^{\text{text}}) \in \mathbb{R}^d$.

B.2 Hypersphere Collapse Prevention

Hypersphere collapse is a trivial optimum of the Eq. 6 objective. When all feature vectors are mapped to the center θ , the radius of the hypersphere becomes zero. We first prove that when all feature extractors are constant, the OCC model is prone to hypersphere collapse. Then, we show that incorporating contrastive learning can prevent this hypersphere collapse.

PROPOSITION B.1. *If the feature extractor is constant, all extracted $\tilde{\mathbf{h}}_i$ and $\tilde{\mathbf{z}}_i$ are equal to θ . In this case, $\mathcal{L}_{\text{OCC}}$ attains its minimum value of 0, leading to hypersphere collapse.*

PROOF. $\mathcal{L}_{\text{OCC}}$ is the squared ℓ_2 norm of the distance between $\tilde{\mathbf{h}}_i, \tilde{\mathbf{z}}_i$ and θ , when $\tilde{\mathbf{h}}_i = \theta, \tilde{\mathbf{z}}_i = \theta$, all embeddings converge to the same point θ , making the distance between $\tilde{\mathbf{h}}_i, \tilde{\mathbf{z}}_i$ and θ equal to 0. Thus, $\mathcal{L}_{\text{OCC}} = 0$. $\square$

In contrast, we combine the OCC loss of S²MOE-F with the contrastive learning loss. The following proof demonstrates that setting a parameter does not minimize the comprehensive loss, thereby helping to avoid hypersphere collapse.

PROPOSITION B.2. *If a certain parameter setting exists, then under the GNN and LLM settings, the constant feature extractor with $\tilde{\mathbf{h}}_i$ and $\tilde{\mathbf{z}}_i = \theta$ cannot make loss equal to 0.*

PROOF. When $\tilde{\mathbf{h}}_i = \theta, \tilde{\mathbf{z}}_i = \theta$, the similarity between $\tilde{\mathbf{h}}_i$ and $\tilde{\mathbf{z}}_i$ is 1, that is $\cos(\tilde{\mathbf{h}}_i, \tilde{\mathbf{z}}_i) = 1$. In this case, the maximum value of contrastive loss is $N \log N$, that is, $\mathcal{L}_{\text{InfoNCE}} = N \log N$, at the same time $\mathcal{L}_{\text{OCC}} = 0$. When integrating the two losses, a parameter μ is introduced so that the loss is $\mu N \log N$, effectively preventing hypersphere collapse. Furthermore, integration of GNN and LLM within the OCC module enhances the diversity of news information and reduces the noise that might arise from a single view. Contrastive learning can also align embeddings across GNN and LLM, and adjust the embedding into a common pattern, making the embeddings more uniform and preventing news from concentrating on the hypersphere surface. $\square$

Proposition B.1 and B.2 demonstrate that our method is not prone to hypersphere collapse. That $\mathcal{L}_{\text{InfoNCE}} < N \log N$ in Proposition 2 can be verified by training the model and evaluating the loss in the training data. In all scenarios we handle, $\mathcal{L}_{\text{InfoNCE}} < N \log N$ is satisfied after training. By combining $\mathcal{L}_{\text{OCC}}$ with $\mathcal{L}_{\text{InfoNCE}}$, our method avoids the degenerated solution where all embeddings collapse to a single point in a high-dimensional space.

B.3 Time Complexity Analysis

For each news item d_i , the overall time complexity of S²MOE-F is $O(|\mathcal{E}| + |\mathcal{V}| + K^2d + Nd)$, where $|\mathcal{V}|$ and $|\mathcal{E}|$ denote the numbers of nodes and edges in the user-news graph, K is the number of sub-embeddings, d is the embedding dimension, and N is the

batch size. Specifically, the GNN expert requires $O(|\mathcal{E}|)$ for message passing, while the LLM expert costs $O(|\mathcal{V}|)$ per batch for textual encoding. The Disentangled Masked Transformer introduces $O(K^2d)$ due to cross-view attention among K sub-embeddings. The OCC and InfoNCE objectives incur $O(Nd)$, and the covariance loss adds a minor $O(Nd^2)$ term. The RL-based router and pseudo-labeling components have lightweight costs of $O(N)$ and $O(N \log N)$, respectively. Overall, the per-epoch training complexity is $O(|\mathcal{E}| + |\mathcal{V}| + K^2d + Nd)$, which scales linearly with the data size. During inference, the complexity reduces to $O(|\mathcal{E}| + |\mathcal{V}| + K^2d)$ since RL routing and pseudo-labeling are omitted.

B.4 The Pseudocode of Proposed Algorithms

In this section, we present the pseudocode of the proposed method S²MOE-F in Algorithm 1.

C Experiment

C.1 Experimental Setting

C.1.1 Baseline. We compare our method with a series of baselines, we list the details of our baselines as follows:

BERT [7] is a pre-trained bidirectional transformer, by fine-tuning on fake news datasets, it captures contextualized semantic representations of news content for detection.

XLNet [54] detects fake news by maximizing the expected likelihood over all possible permutations of the factorization order, while incorporating contrastive learning to enable effective self-supervised representation learning.

OCC [34] describes an one-class classification approach designed to identify fake news data that deviate from the true news.

UFD [53] leverages a generative model to learn the latent features of news without labels.

GCNFN [26] encodes news propagation graphs with GCN while incorporating user features derived from profile information and comment text embeddings.

GNN-CL [11] combines graph neural networks with continuous learning to achieve fake news detection.

GNN [49] captures relationships in the news to build a graph structure, using these features to distinguish true or fake news.

GAMC [59] detects fake news by augmenting news diffusion graphs and learning representations through a graph autoencoder optimized with both reconstruction loss and contrastive loss.

ARG [12] employs LLMs to generate multi-perspective reasoning evidence, encodes both news and evidence with BERT, and integrates them through interaction and evaluation modules to predict news authenticity.

InfOp [47] detects fake news by continually improving social context representations using graph neural networks.

MFCL [6] detects fake news by learning robust multi-modal representations through cross-modal and propagation network contrastive learning.

C.1.2 Detailed Settings. All experiments are implemented on Python 3.10.0, CUDA 11.8.1, a 10-core CPU, 1 NVIDIA GeForce RTX 4080, PyTorch 2.3.0, and 64GB RAM. We perform grid searches for the key hyperparameters of S²MOE-F and all baselines. In Section 4.5, we perform a grid search and analyze the parameters of S²MOE-F,

including λ , μ , γ , and batch size. For the baselines, we also grid searched important parameters, such as the embedding dimension, epochs and learning rate (LR).

C.2 Evaluation of Robustness to Unseen Publishers

We conduct an experiment on two datasets, GossipCop and Weibo-21. For GossipCop, training is performed on four major publishers, and two small publishers are exclusively reserved for testing. For Weibo-21, the model is trained on seven major publishers, while three small publishers are only included at the test stage. This experimental setting is designed to examine the model’s ability to generalize beyond dominant publisher distributions and to assess

Algorithm 1: S^2 MOE-F

Input : Labeled dataset $\mathcal{D}_l$; Unlabeled dataset $\mathcal{D}_u$; GNN expert $\mathcal{H}_{\text{GNN}}$; LLM expert $\mathcal{H}_{\text{LLM}}$; Iterations T ; Number of sub-embeddings K .

Output : Trained S^2 MOE-F model; misinformation scores s_i for test news d_i .

```

1 Initialize parameters of  $\mathcal{H}_{\text{GNN}}$ ,  $\mathcal{H}_{\text{LLM}}$ , Disentangled Masked
  Transformer, and RL-router;
2 for  $t \leftarrow 1$  to  $T$  do
3   for each  $d_i \in \mathcal{D}_l \cup \mathcal{D}_u$  do
4     Generate structural embedding  $\mathbf{h}_i \leftarrow \mathcal{H}_{\text{GNN}}(d_i)$ ;
5     Generate semantic embedding  $\mathbf{z}_i \leftarrow \mathcal{H}_{\text{LLM}}(d_i)$ ;
6     Partition embeddings into  $K$  sub-embeddings
        $\{\mathbf{h}_i^k, \mathbf{z}_i^k\}_{k=1}^K$ ;
7     Apply Disentangled Masked Transformer to obtain
       refined embeddings  $\tilde{\mathbf{h}}_i, \tilde{\mathbf{z}}_i$ ;
8     if  $d_i \in \mathcal{D}_u$  then
9       Compute confidence score  $\text{Conf}(\tilde{\mathbf{h}}_i, \tilde{\mathbf{z}}_i)$  (Eq. 9);
10      Assign pseudo-label  $y_i$  if confidence is high;
11      Add high-confidence pseudo-labeled samples to
         $\mathcal{D}_l$ ;
12    Update RL-router weights  $w_{\text{GNN}}, w_{\text{LLM}}$  using
      batch-aware reward (Sec. 3.5);
13    Compute OCC loss  $\mathcal{L}_{\text{OCC}}$  (Eq. 6);
14    Compute InfoNCE loss  $\mathcal{L}_{\text{InfoNCE}}$  (Eq. 7);
15    Compute covariance loss  $\mathcal{L}_{\text{Cov}}$ ;
16    Update model parameters by minimizing
       $\mathcal{L} = \lambda \mathcal{L}_{\text{OCC}} + \mu \mathcal{L}_{\text{InfoNCE}} + \gamma \mathcal{L}_{\text{Cov}}$  (Eq. 8);
17 for each test news  $d_i$  do
18   Generate embeddings  $\tilde{\mathbf{h}}_i, \tilde{\mathbf{z}}_i$ ;
19   Apply pseudo-labeling to select reliable pseudo-true
     examples;
20   Update RL-router weights  $w_{\text{GNN}}, w_{\text{LLM}}$  using
     pseudo-true samples;
21   Compute misinformation score
      $s_i = w_{\text{GNN}} \|\tilde{\mathbf{h}}_i - \boldsymbol{\theta}\|_2 + w_{\text{LLM}} \|\tilde{\mathbf{z}}_i - \boldsymbol{\theta}\|_2$  (Eq. 13);
22 return Trained  $S^2$ MOE-F model and misinformation scores  $s_i$ ;

```

Table 3: Performance Comparison under Standard Setting and Unseen Small Publishers.

Datasets	Setting	AUC $\uparrow$	F1 $\uparrow$
GossipCop	Unseen Small Publishers	78.40 ± 3.51	78.63 ± 5.57
	Standard-setting	80.13 ± 2.06	79.36 ± 2.14
Weibo-21	Unseen Small Publishers	76.05 ± 2.98	74.57 ± 4.18
	Standard-setting	76.85 ± 2.17	74.25 ± 2.39

Table 4: Comparison of RL Routing and Fixed-Weight in Inference Stage.

Datasets	Methods	AUC $\uparrow$	F1 $\uparrow$
COVID-19	Fixed-Weights	84.38 ± 2.51	77.36 ± 3.42
	S^2 MOE-F ^L	84.21 ± 3.35	78.79 ± 3.22
GossipCop	Fixed-Weights	80.11 ± 2.45	79.11 ± 3.70
	S^2 MOE-F ^L	80.13 ± 2.06	79.36 ± 2.14
PolitiFact	Fixed-Weights	79.21 ± 2.98	76.24 ± 2.15
	S^2 MOE-F ^L	80.03 ± 1.59	77.10 ± 1.37

its robustness under publisher distribution shifts, as it intentionally creates a clear distribution gap between the training and test datasets by strictly separating major and minor publishers. In Table 3, we can see the performance changes are relatively small with no obvious decline in core evaluation indicators, showing that our OCC learns a robust center of true news rather than being tied to major publishers, which indicates its effectiveness when encountering previously unseen small publishers.

C.3 Effectiveness of RL-based Routing during Inference

To evaluate the effectiveness of the proposed RL-based routing strategy during inference, we conduct experiments on three datasets, including COVID-19, GossipCop, and PolitiFact. In these three datasets, we compared our RL-router against the conventional fixed weights strategy during the inference stage to conduct a direct and rigorous performance comparison. As reported in Table 4, the RL router achieves comparable or slightly improved performance in terms of both AUC and F1 score across all evaluated benchmarks. In particular, on the COVID-19 dataset, the RL router shows a modest yet stable improvement over the fixed-weight scheme, suggesting that adaptive routing can better accommodate variations in inference-time data distributions. Similar trends are observed on GossipCop and PolitiFact, where the RL-based routing strategy consistently outperforms the fixed-weight baseline with marginal yet consistent gains in the core AUC and F1 metrics. These results indicate that dynamically adjusting routing decisions at inference time remains beneficial, and provides a more flexible and robust strategy than relying on static weights that are pre-determined and unchangeable during inference.